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Hierarchical Algorithms
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Exercise sheet 11, Solution hints
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Homework:

Exercise 1: Let

$$A := \begin{pmatrix} 1 & 0 & 2 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 2 & 8 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

with the singular values $\sigma_1 \approx 8.54777, \sigma_2 \approx 4.09514, \sigma_3 \approx 1.07955, \sigma_4 = 0$.

a) Compute (by hand) an $\mathcal{R}(3)$ representation $A = UV^T$ with $U, V \in \mathbb{R}^{4\times 3}$ using the cross approximation algorithm.

b) Determine permutation matrices $P, Q \in \mathbb{R}^{4\times 4}$ and lower triangular matrices $L, R^T \in \mathbb{R}^{4\times 3}$ such that there holds $PAQ = LR$.

c) Determine a suitable index set $J = \{j_1, j_2, j_3\} \subset I := \{1, 2, 3, 4\}$ such that there holds $A = A|_{I\times J} \left(A|_{J\times J}\right)^{-1} A|_{J\times I}$.

d) Compute rank-2 approximations R of A with the help of cross approximation

- with complete pivot search,
- with adaptive pivot choice, both times beginning at row 4.

Compute the respective approximation error $\|A - R\|_F$ and compare it to the error $\|A - R_{best}\|_F$ of the rank-2 best approximation R_{best} of A.

Solution hints:

a) For $\ell = 1$ we choose $a_{33} = 8$ as the pivot. The 3rd row and the 3rd column of A are already given, hence

$$u_1 = \begin{pmatrix} 2 \\ 0 \\ 8 \\ 0 \end{pmatrix}, \; v_1 = \frac{1}{8} \cdot \begin{pmatrix} 0 \\ 2 \\ 8 \\ 0 \end{pmatrix} \text{ and } u_1 v_1^T = \begin{pmatrix} 0 & 0.5 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

We update A by computing

$$A \leftarrow A - u_1 v_1^T = \begin{pmatrix} 1 & -0.5 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

For $\ell = 2$ we choose $a_{22} = 4$ as the pivot, hence

$$u_2 = \begin{pmatrix} -0.5 \\ 4 \\ 0 \\ 1 \end{pmatrix}, \; v_2 = \frac{1}{4} \cdot \begin{pmatrix} 1 \\ 4 \\ 0 \\ 0 \end{pmatrix} \text{ and } u_2 v_2^T = \begin{pmatrix} -0.125 & -0.5 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.25 & 1 & 0 & 0 \end{pmatrix}.$$

We update A by computing

$$A \leftarrow A - u_2 v_2^T = \begin{pmatrix} 1.125 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -0.25 & 0 & 0 & 0 \end{pmatrix}.$$

For $\ell = 3$ we choose $a_{11} = 1.125$ as the pivot, hence

$$u_3 = \begin{pmatrix} 1.125 \\ 0 \\ 0 \\ -0.25 \end{pmatrix}, \; v_3 = \frac{8}{9} \cdot \begin{pmatrix} 1.125 \\ 0 \\ 0 \\ 0 \end{pmatrix} \text{ and } u_3 v_3^T = \begin{pmatrix} 1.125 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -0.25 & 0 & 0 & 0 \end{pmatrix}.$$

The final update of A yields

$$A \leftarrow A - u_3 v_3^T = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Overall we obtain $A = UV^T$ with

$$U = \begin{pmatrix} 2 & -0.5 & 1.125 \\ 0 & 4 & 0 \\ 8 & 0 & 0 \\ 0 & 1 & -0.25 \end{pmatrix} \text{ and } V = \begin{pmatrix} 0 & 0.25 & 1 \\ 0.25 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

b) We use the decomposition $A = UV^T$ from part a) which we obtained using the pivot elements a_{33}, a_{22} und a_{11}. The corresponding column and row permutations

$$P := Q^T := \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

lead to $PAQ^T = PUV^TQ^T =: LR$ with

$$L := PU = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 2 & -0.5 & 1.125 \\ 0 & 1 & -0.25 \end{pmatrix} \text{ and } R^T := QV = \begin{pmatrix} 1 & 0 & 0 \\ 0.25 & 1 & 0 \\ 0 & 0.25 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

An altenative representation is obtained (without the use of part a)) via the computation of an LU-decomposition of A where the last column of L and the last row of R are removed: For $P := Q := I \in \mathbb{R}^{4\times 4}$ (no pivoting is necessary in this case) and

$$L := \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & \frac{1}{4} & \frac{1}{18} \end{pmatrix} \quad \text{as well as } R^T := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 2 & -2 & 9 \\ 0 & 0 & 0 \end{pmatrix}$$

there holds $PAQ = LR$.

c) For $J = \{1, 2, 3\}$ it holds that

$$A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 2 & 8 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & 2 & 8 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & 2 & 8 \\ 0 & 1 & 0 \end{pmatrix} = A|_{I\times J} \left(A|_{J\times J}\right)^{-1} A|_{J\times I}.$$

d) In part a) we carried out a complete pivot search. This results in

$$R = u_1 v_1^T + u_2 v_2^T = \begin{pmatrix} 0 & 0.5 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} -0.125 & -0.5 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.25 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -0.125 & 0 & 2 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 2 & 8 & 0 \\ 0.25 & 1 & 0 & 0 \end{pmatrix}.$$

It follows that

$$\|A - R\|_F = \left\| \begin{pmatrix} 1.125 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -0.25 & 0 & 0 & 0 \end{pmatrix} \right\|_F = \sqrt{(1.125)^2 + (-0.25)^2} = \sqrt{\frac{85}{64}} \approx 1.15244.$$

Further we obtain $\|A - R_{best}\|_F = \sqrt{\sum_{i=3}^{4} \sigma_i^2} = \sigma_3 \approx 1.07955$.

Now we compute a rank-2 approximation R of A using cross approximation with adaptive pivoting where we start with row 4 both times.

For $\ell = 1$ we start with row 4. The 4th row of A is given, there were no previous updates and the largest element by absolute value in this row is a_{42}. The 2nd column of A is also given, there were no previous updates here neither and the largest element by absolute value in this column is a_{22}. Therefore we choose $a_{22} = 4$ as the pivot and set

$$u_1 = \begin{pmatrix} 0 \\ 4 \\ 2 \\ 1 \end{pmatrix} \quad \text{and } v_1 = \frac{1}{4} \cdot \begin{pmatrix} 1 \\ 4 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

For $\ell = 2$ we again start with row 4 and first compute the update from the first step for this row:

$$\begin{aligned} a_{4,:} &\leftarrow a_{4,:} - u_{1_4} v_1^T \\ &= a_{4,:} - 1 \cdot \begin{pmatrix} 0.25 & 1 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -0.25 & 0 & 0 & 0 \end{pmatrix}, \end{aligned}$$

where u_{1_4} is the 4th element of the vector u_1. The largest element by absolute value in this row is a_{41}. So we compute the update for the first column:

$$\begin{aligned} a_{:,1} &\leftarrow a_{:,1} - u_1 v_{1_1} \\ &= a_{:,1} - \frac{1}{4} \cdot u_1 \\ &= \begin{pmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ -\frac{1}{4} \end{pmatrix}. \end{aligned}$$

The largest element by absolute value in this column is a_{11}. So we choose $a_{11} = 1$ as the second pivot element. The new first column of A is already computed and for the new first row of A we obtain

$$a_{1,:} = a_{1,:} + u_{1_1} v_1^T = \begin{pmatrix} 1 & 0 & 2 & 0 \end{pmatrix} - 0 \cdot v_1^T = \begin{pmatrix} 1 & 0 & 2 & 0 \end{pmatrix}.$$

It follows that

$$u_2 = \begin{pmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ -\frac{1}{4} \end{pmatrix} \text{ and } v_2 = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix}.$$

Using

$$U = \begin{pmatrix} 0 & 1 \\ 4 & 0 \\ 2 & -0.5 \\ 1 & -0.25 \end{pmatrix} \text{ and } V = \begin{pmatrix} 0.25 & 1 \\ 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$$

yields

$$R = UV^T = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 1 & -0.5 & 0 \end{pmatrix}.$$

Therefore

$$\begin{aligned} \|A - R\|_F &= \left\| \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0.5 & 0 \end{pmatrix} \right\|_F \\ &= \sqrt{9^2 + 0.5^2} \approx 9.0139. \end{aligned}$$

The cross approximation with complete pivot search yields a much better result for this example compared to the cross approximation with adaptive pivoting.

Exercises (in class):

Exercise 2: Implement the adaptive cross approximation by completing the following functions in *aca.c*:

- ```
  double compute_entry_aca(pcrkmatrix r, int d, int k, int row, int col,
            const double *x, const double *y,
            double (*test_function)(int d, const double *x, const double *y));
  ```

- ```
  prkmatrix aca_rkmatrix(int d, int k, int rows, int cols,
            const double *nodes_x, const double *nodes_y,
            double (*test_function)(int d, const double *x, const double *y));
  ```

Let $rows, cols, d \in \mathbb{N}$ and let $rows+cols$ points $x_1, \ldots, x_{rows}, y_1, \ldots, y_{cols} \in \mathbb{R}^d$ be given. Further let $g\colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a kernel function. We define the matrix $A \in \mathbb{R}^{rows \times cols}$ by $A_{i,j} := g(x_i, y_j)$ for all $i \in \{1, \ldots, rows\}, j \in \{1, \ldots, cols\}$. The first function should compute

$$test_function(d, x, y) - \sum_{i=0}^{k-1} r \to a[i * r \to rows + row] \quad * \quad r \to b[i * r \to cols + col]$$

(compare to step 5 in the cross approximation algorithm in the lecture notes).

The second functions generates an $\mathcal{R}(\min\{k, rows, cols\}, rows, cols)$ matrix by adaptive cross approximation which approximates the matrix A.

Test your functions using *example11.c*.

Note: *example11.c* corresponds to example program *example09.c* where the call of the function

```
fill_kernelinterpolation_supermatrix(s, d, degree, nodes_x,
                                     test_function_log, 0, 0, nd);
```

was replaced by the call of the function

```
fill_kernelaca_supermatrix(s, d, rank, nodes_x, test_function_log, 0, 0, nd);
```

The is already implemented in *supermatrix.c* and computes $\mathcal{R}$(k)-matrices in admissible blocks with adaptive cross approximation

```
s->r = aca_rkmatrix(d, k, s->rows, s->cols, nodes_row, nodes_col, test_function);
```

whereas before interpolation (`s->r = interpolate_rkmatrix(d, ...);`) was used (compare Exercise 3 on Sheet 06).

Solution hints:

Exercise 3: Modify the function for the adaptive cross approximation (*aca_rkmatrix* and *fill_kernelaca_supermatrix* from the previous exercise) such that in addition to the (maximal) rank k a δ is passed to the function which is used in the heuristic stopping criterion

$$\|u_\ell\|_2 \|v_\ell\|_2 \leq \delta \|u_0\|_2 \|v_0\|_2,$$

i. e. an `rkmatrix` with r→ $kt = \min\{k, \ell \mid \ell \text{ satisfies the above inequality}\}$ is computed.
To this end complete the function *aca_delta_rkmatrix* in *aca.c* and the function *fill_kernelaca_delta_supermatrix* in *supermatrix.c*.

Hint: You can also use *example11.c* to test this function.

Note: In Exercise 4 on Sheet 09, an $\mathcal{H}$-Matrix with fixed rank k was set up and then recompressed to an accuracy ε. Alternatively, an $\mathcal{H}$-matrix can be set up directly with adaptive ranks via the above heuristic termination criterion with the parameter δ by the function *fill_kernelaca_delta_supermatrix*.

Solution hints: